Static-Shield

Passive Mosquito Protection System No Electricity • No Chemicals • No Electronics

Innovative Pest Control Technology

2025

What is Static-Shield?

Static-Shield is a **passive protection barrier** against mosquitoes that works completely without electricity, poison, or electronics.

It consists of a beam with mineral dust that becomes electrically charged through friction, creating an invisible zone that disrupts mosquitoes.

- No Electricity
- No Chemicals
- No Electronics
- Cost-Effective

Core Principle: Triboelectric charging creates mechanical and electrical disruption for mosquitoes

How It Works

1. Dust Preparation

Fill the Channel

Channel filled with **diatomaceous earth and lime/clay**. Acts like tiny razor blades on insects.

2. Friction Activation

Generate Charge

Band moved briefly creates **triboelectric charge** - dust becomes electrically charged.

3. Active Zone

Create Protection

Charged dust forms active zone above channel that disrupts mosquitoes.

4. Mosquito Disruption

Protection Effect

Mosquitoes are **mechanically damaged and electrically irritated**, disrupting orientation.

Result: Mosquitoes avoid the area or become weakened - reducing bites and disease risk

What Happens to the Mosquito?

1. **Mosquito Approaches**
   Area above dust channel is electrically active from friction charging.

2. **Particles Adhere**
   Charged dust particles stick to wings, legs, and body surface.

3. **Mechanical Effect**
   Diatomaceous earth destroys wax layer → mosquito dries out.

4. **Electrical Effect**
   Charge disrupts orientation → mosquito becomes irritated or flies away.

Result: Mosquitoes avoid area → fewer bites → reduced disease risk

Key Advantages

Core Benefits

No Electricity: Works completely without power supply

No Chemicals: Safe for humans, animals, and environment

No Electronics: No complex technology or maintenance

Passive Protection: Works 24/7 without supervision

Practical Benefits

Extremely Low Cost: ~€13.50 per module material cost

Local Production: Can be built with local materials

Immediate Use: Ready to deploy after assembly

Universal Application: Works in any climate and location

Target Groups & Applications

Sleeping Areas

- **Beds and mats:** Protection around sleeping areas in open rooms or tents
- **Huts and shelters:** Perimeter protection for temporary accommodations
- **Verandas:** Coverage for outdoor sleeping and resting areas

Installation Sites

- **House edges:** Along walls, fences, or entrances
- **Villages and camps:** Local production and deployment in rural regions
- **Water sources:** Around wells and water collection points

Target Organizations

- **NGOs:** Projects without electricity, focus on health and self-construction
- **Development projects:** Community-based mosquito control initiatives
- **Self-build initiatives:** Local manufacturing and implementation

Ideal Regions

- **Rural areas:** Remote locations without reliable electricity
- **Tropical zones:** High mosquito activity regions
- **Health-critical areas:** Malaria and dengue endemic regions

Perfect for: Communities needing affordable, sustainable mosquito protection without infrastructure dependencies

Materials & Components

Support Beam

Material: Wood, metal or stable plastic

Dimensions: 1m length, 5×5cm cross-section

Function: Mechanical base for all components

Dust Channel

Shape: U-shaped groove with edge lips

Size: 3-4cm wide, 1-2cm deep

Fill: 70% diatomaceous earth, 30% lime/clay

Grid Cover

Material: Fine mesh, plastic or metal

Function: Holds dust, allows air and charge through

Purpose: Prevents dust from blowing away

Friction Element

Options: Rubber band, brush strip, or fabric

Function: Creates triboelectric charge through movement

Usage: 10-20 seconds daily activation

Total Cost: Approximately €13.50 per module - extremely cost-effective protection

Assembly & Installation

Assembly Steps

- **Cut beam:** 1m length, 5×5cm cross-section
- **Create channel:** U-shaped groove 3-4cm wide, 1-2cm deep
- **Mount friction element:** Band or brush above channel
- **Fill dust:** 70% diatomaceous earth, 30% lime/clay

Installation Guide

- **Height:** 0.5-1.5m above ground (mosquito flight level)
- **Ring setup:** 6 modules for standard protection circle
- **Spacing:** Max 20cm gaps between modules
- **Mounting:** Stable supports, posts, or brackets

Ready to Use: Activate by moving friction element 10-20 seconds daily before mosquito activity

Operation & Maintenance

Daily Operation

- **Evening activation:** 10-20 seconds per module before mosquito activity
- **Wind movement:** Automatic activation or manual movement
- **Manual operation:** Brief friction element movement creates charge
- **Protection duration:** Active for several hours after activation

Maintenance

- **Dust refill:** Check every 1-2 weeks, refill as needed
- **After rain:** Check dust level and refill if washed away
- **Cleaning:** Clean friction element when dirty or stuck
- **Grid check:** Ensure mesh cover remains secure and intact

Simple Maintenance: Minimal upkeep required - just dust refilling and occasional cleaning

Ring System for Maximum Protection

Standard Ring

- **6 modules à 1m:** Circular arrangement around target object
- **Distance:** Max 20cm between modules for seamless effect
- **Height:** 0.5-1.2m above ground optimal

Extended Ring

- **8-12 modules:** For larger areas and buildings
- **Shape:** Circular or square as needed
- **Application:** Huts, seating areas, community spaces

Operation & Activation

- **Daily:** 10-20 seconds per module activation
- **Wind:** Automatic movement or manual activation

Maintenance

- **Refill:** Check dust every 1-2 weeks
- **Cleaning:** Clean friction element when dirty

Result: Charged dust particles create an irritating zone - mosquitoes avoid the area or become weakened

Cost Overview

€13.50 / $15.93

Material Cost per Module (EUR/USD)

Structural Materials

€8.50 / $10.03

Wood beam, channel, grid cover, and mounting hardware

Active Materials

€3.00 / $3.54

Diatomaceous earth, lime/clay mixture, and friction element

Assembly & Tools

€2.00 / $2.36

Screws, adhesive, and basic assembly materials

Economic Impact

Significant

Reduces healthcare costs and improves quality of life in affected communities

Investment Return: One-time cost provides years of protection - no ongoing electricity or chemical costs (Exchange rate: 1 EUR = 1.18 USD)

Implementation & Impact

Implementation Scenarios

- **Rural households:** 6-module ring around sleeping areas
- **Community centers:** Extended 8-12 module protection
- **Health clinics:** Perimeter protection for patient areas
- **Schools:** Playground and classroom protection

Expected Impact

- **Bite reduction:** 60-80% fewer mosquito encounters
- **Health improvement:** Reduced malaria and dengue transmission
- **Better sleep:** Undisturbed rest without chemical exposure
- **Sustainability:** Long-term protection with minimal maintenance

Ready to Deploy

Static-Shield offers immediate, affordable mosquito protection for communities worldwide. Start building your protection system today with locally available materials.